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State-Based Process Monitoring in Traditional Event Logs for Concept Drift Detection

Bachelor Thesis

presented by

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Acknowledgments

If you want to incorporate it, keep in mind that, according to the official regulations, *independent work* is a very important aspect of writing a thesis. Thus, you can be grateful for the emotional support by your favorite pet, yet we recommend you to be careful if you want to write anything more specific. In general, we suggest to omit this section.

Abstract

The abstract should be a concise summary of what you have done. It should cover, in roughly 1 to 2 sentences per topic:

- The domain, i.e., explain the application domain in which your thesis is applicable.
- The problem, i.e., motivate what the problem is you are solving.
- The method, i.e., describe the method you have proposed, and, compare it, briefly & high level to other approaches. In particular, focus on the benefits of your approach versus the other approach(es)
- The evaluation, i.e., how did you evaluate your method, what results did you obtain and what do the results indicate?

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1 Introduction

First of all, we encourage you to read Prof. van der Aalst’s guide on how to write papers [8].

In this section, you introduce the work to your audience. Even though this is the first chapter of your thesis, you should not necessarily write this chapter first. It is often easier to write this chapter at the end of the thesis writing. In this part of your introduction, i.e., the part before the Motivation section (Section 1.1), you write a paragraph (not just a sentence) for each of the bullet points mentioned in the abstract:

- Domain, i.e., explain the application domain in which your thesis is applicable.
- Problem, i.e., motivate what the problem is you are solving.
- Method, i.e., describe the method you have proposed, and, compare it, briefly & high level to other approaches. In particular, focus on the benefits of your approach versus the other approach(es)
- Evaluation, i.e., how did you evaluate your method and what do the results indicate?

Optionally, you can finalize this part with a little overview of what is still to come in this chapter: “The remainder of this chapter is structured as follows. In Section 1.1, we motivate the need for ... In Section 1.2, we provide a formal problem statement. ...”

Before we dive into the motivation section, a few general guidelines for writing:

1. Write *active* and in the *present tense*.
2. Avoid “optionality” as much as possible, i.e., usage of the following words should (pun intended) be minimized:
 - could
 - would
 - should
 - may
 - might
 - can
 - will
 - ...

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Good: We present a method that ...

Bad: We will present a method that ...

Good: We use function f and compute its inverse.

Bad: We can use the function f and will compute its inverse.

3. Do not use `\\` to generate line breaks, but an empty line
4. Position the caption of a Figure below the figure.
5. Position the caption of a Table above the table.
6. Write Section Headings and Titles Like This
Good: Approximation Bias in Unstable Systems
Bad: Approximation bias in unstable systems
7. Be consistent in your references.
 - Make sure that the name of the same author is always the same (using DBLP helps for this)
 - Titles should be consistent. Either use the style previously described, i.e., Approximation Bias in Unstable Systems or Approximation bias in unstable systems (this is allowed in the references, opposed to your own section headings and titles, however, *BE CONSISTENT*).

1.1 Motivation

Played-out event logs do not change, but real processes do. Following van der Aalst [7], the situation in which a process is changing while being analyzed is called a concept drift. For example, two activities are concurrent at the start of an event log but then become sequential. According to van der Aalst [7], examples to main triggers of concept drifts are variation like more demand in December, seasonal effects, or changed market conditions. Classical process mining techniques mostly assume that processes at the start and end of the log are the same (Bose et al. [3]). In case of a drift, process models or key performance indicators (KPI) over the entire log though mixes two different processes. The process owner therefore must detect, if and when the process changed and what changed.

A drift does not restrict itself to the order of activities. Processes run together with a context, which is in analysis sometimes ignored. As described by van der Aalst [7], process context covers how cases influence each other like competing for the same resources. A single case is therefore not sufficient to explain the observed behavior alone. Thus, for successful explanation of the observed behavior and changes, dependencies inside the cases as well as between the cases and dependencies with the resources need to be analyzed. This leads to the terms of intra-case dependencies and inter-case dependencies introduced by Senderovich et al. [6]. This thesis takes on the terminology and considers a process from three perspectives:

- *Intra-case perspective:* The history of a single case up to a specific time, covering which activities happened in which order and how often. Drift example: A new trace variant dominates.
- *Resource perspective:* The working style of resources like their workload, service times, handover, responsible activities. Drift example: A resource no longer exists and other resources take over the work.
- *Inter-case perspective:* The current situation of all running cases of the process, covering arrivals, waiting times, case attributes. Drift example: The mean waiting time between consecutive events of a case increases.

From all three perspectives concept drifts can take place, but they can also influence each other. E.g., a new trace variant having a frequent activity becomes dominant (intra-case perspective), the responsible resource of that activity has more work to do (resource perspective), waiting times in the entire process are affected (inter-case perspective).

Existing concept drift detection approaches cover this picture only partially:

- *Control-flow focus:* Most of the work on drift detection focuses on control-flow. Out of 38 papers in the survey of Sato et al. [5], only six focus on the time perspective, three on data perspective, and none on the resource perspective. In our framework, the time and data perspectives fall under the inter-case perspective, and the resource perspective corresponds to our resource perspective. A concept drift causing changes in time or resources perspectives stays undetected with those approaches.
- *Single KPIs:* Approaches beyond control-flow often track single KPIs. Adams et al. [1] turn each KPI chosen by the user into a time series over time windows and detect drifts in each series separately. A drift therefore shows up as separate change points in several KPIs. This approach does not deliver a compact description of the overall process situation. For steady-state detection, Kraus et al. [4] show that combining complementary process characteristics is more accurate than relying on a single one.
- *Missing test data:* Sato et al. [5] argue that real-world logs are unsuitable for measuring detection accuracy, since their drift points are unknown. They find only three public synthetic datasets with injected drifts. These datasets cover only the control-flow and time perspectives and do not mix different drift types. None of them contains a resource drift or a change in case arrivals.

Figure 1.1 shows this gap using a synthetic event log generated using Rheon. Rheon is our generator for synthetic event logs with injected concept drifts in all three perspectives (Chapter 4). This log has 300,299 cases with 1,946,540 events, seven activities, and seven resources. Its cases arrive between January 1 and December 31, 2020. On April 19, 2020, the mean waiting time between consecutive events of a case jumps from 14.2 minutes to 45 minutes. The activities, the control-flow, the resources, the resource-activity

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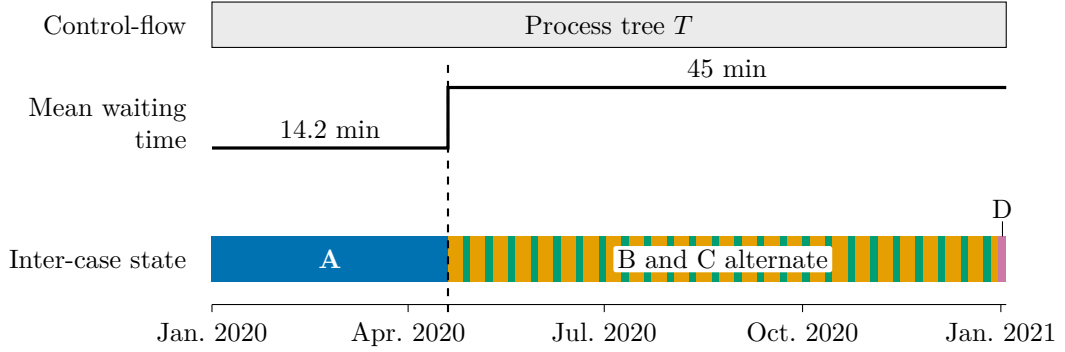


Figure 1.1: Visualization of the synthetic Rheon log generated with mean waiting time drift. At the drift point (dashed), the mean waiting time between consecutive events of a case rises from 14.2 to 45 min, while all cases still follow process tree T . The inter-case state of each 12-hour window switches exactly there from A to an alternation of B and C, D marks the run-out of the log after the last case arrivals. The alternation of B and C is drawn schematically.

dominances remain unchanged. A detector only comparing relationships between activities will see before and after the drift point the same control-flows and will be unable to detect a change.

The state-based approach of this thesis aims to detect drifts in all three perspectives and to show what changed. Instead of separate change points per KPI, our states combine many features into one compact description, and Rheon provides test logs with known drift points in all three perspectives. In Berti et al. [2], execution states from object-centric event logs are computed for objects. We adapt the approach from objects to cases in our case-centric framework and define our intra-case states. As an extension to their work, our approach also takes into account resource and inter-case perspectives and adapts the idea to them.

Overall, we compute states from an event log by extracting features, compressing, and clustering. The input of our approach is an event log, and the output consists of intra-case, resource, and inter-case states whose changes over time indicate drifts. In Figure 1.1 one can visually see the general idea of our approach for the inter-case perspective. We describe each 12-hour window by features such as the number of active cases, arrivals, completions, and the mean waiting time between consecutive events of a case. Following the execution states of Berti et al., a principal component analysis (PCA) compresses these features. A self-organizing map (SOM) with 2×2 is trained on these compressed feature vectors and its cells assign each window one of four states.

The pipeline is unsupervised and does not see the drift point or the drift type. As seen in Figure 1.1 all 219 windows before the drift point fall into state A. From the drift point until December 31, the windows alternate between states B and C. The last seven windows, in which the log runs out, form state D. The first principal component explains 51% of the variance and is dominated by the mean waiting time between consecutive

events of a case. It therefore also shows what changed: after the drift point, more time passes between the events of a case.

1.2 Problem Statement

1.3 Research Questions

The research questions you pose, are questions you need to (largely) answer in order to solve the problem. Basically, the combined set of answers to the questions you pose, allow you to solve the research problem.

From the book of Justin Zobel, “Writing for Computer Science” [9] (which we highly recommend you to read before writing):

A hypothesis or research question should be specific and precise, and should be unambiguous; the more loosely a concept is defined, the more easily it will satisfy many needs simultaneously, even when these needs are contradictory. And it is important to state what is not being proposed—what the limits on the conclusions will be.

In the context of our example, we can define many relevant research questions:

1. What are typical prominent noise patterns in real event data?
2. How to detect noise patterns in event data?
3. How to detect infrequent behavior in event data?
4. How to detect contextually different executions of the same activity?
5. How to balance between detected noise patterns, infrequent behavior and imprecise labels?
6. ...

Typically, your thesis consists of 3-5 research questions (often multiple more questions can be defined).

1.4 Research Goals

In this section, you present your research goals. A research goal is a concise statement that s.t., achieving the goal helps you in (partially) solve a research question. Hence, the research questions and goals are often very related. Furthermore, it should be obvious for the reader what goal answers what problem.

In the context of our previously mentioned questions, some example goals are:

- Conduct a systematic literature review on noise patterns in real event data.

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- Design a noise detection algorithm.
- Formalize the notion of incompleteness in an event log.
- ...

1.5 Contributions

In this section, you list the contributions that your thesis makes to our wonderful world (of science). Again, there is a strong link to the previous section. Usually, you have achieved your research goals. Hence, the contributions are concrete statements of the goals you have achieved. Additionally, your evaluation (most likely also stressed as a goal) is a contribution. Any implementation or prototype can also be quantified as a contribution.

Some examples:

- A systematic literature review covering 35 articles on noise patterns in real event data
- A noise detection algorithm based on dynamic programming and symbolic linking
- ...

1.6 Thesis Structure

In this section, you outline the remainder of the thesis.

The remainder of this thesis is structured as follows. In Chapter 2, we discuss related work. In Chapter 3, we present basic mathematical preliminaries, used throughout the thesis.

2 Related Work

In this section, you are going to list all the related work. In this template, the related work section is located directly after the introduction chapter, however, this is not always the best/most logical location. As a general guideline, the position of the related work section can be:

- *Directly after the introduction*; Position the related work section here, in case you do not need to refer to any detailed (mathematical) concepts. Furthermore, you do not need to refer to any part of your solution. Positioning the related work section here is most commonly done if you are really uncovering “new terrain”, i.e., you are doing something *very novel*.
- *Directly after preliminaries*; In this scenario, you swap the related work and preliminaries, i.e., w.r.t. this thesis template. Position this here when you do need to refer to some of the (mathematical) concepts (typically presented in the preliminaries), yet, you do not need to use any properties of your own solution when comparing your work to the related literature.
- *Directly before the Conclusion (typically after the Evaluation)*; Position the related work section here, in case you need to compare the related work against properties of your algorithm and/or the results you have obtained.

In some cases, you can introduce sub-sections in this chapter, i.e., if a division of the related work eases the readability.

Note that the related work is not intended for you to “Convince your supervisors that you have studied a lot of papers”. Rather, you simply describe in 1, max 2 lines what the core idea of a paper is, how it relates to your work and why it is different from your work. In some cases, it is possible to discuss the aforementioned points for a selection of papers rather than a single paper. Try to present the paper along logical lines, i.e., define categories of work and introduce subsections for this. Never present a long 2-page list of randomly/chronologically structured work. Always add a decent structure!

3 Preliminaries

In the preliminaries you present well-understood mathematical concepts that you need in your thesis. For example, you can define the natural numbers as $\mathbb{N} := \{1, 2, \dots\}$, and, correspondingly $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. A preliminary notion is either a well-defined commonly understood mathematical notion, e.g., sets, multisets, graphs, sequences, Petri nets, $\dots$, or, it is a concept clearly defined in another paper, i.e., you just adopt the notation (or a slight variation thereof). *Any concept you use should be defined in your thesis.* You should never write: “We use Workflow nets, a definition of these can be found here [X]”. If you use it, explain it.

Concepts that are unique to your approach are not part of the preliminaries, i.e., they are described in the approach section itself.

Some useful tips:

- When introducing a complex concept, use the following structure (always works):
 - Explain the concept informally.
 - Provide a formal definition of the concept.
 - Provide an example, using the formal definition, of the concept.

In your examples, try to be *as visual as you can*, often, an image says more than 5 pages of text.

- use commands, some of which are already provided in the preamble, e.g.,
`\newcommand*{\N}{\mathbb{N}}`

4 Rheon

Placeholder.

Table 4.1: Placeholder table.

Column A	Column B
Placeholder	Placeholder

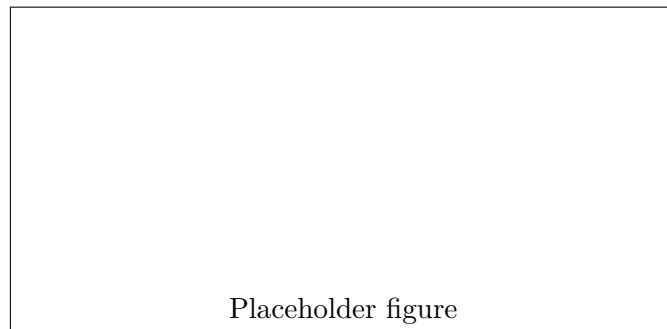


Figure 4.1: Placeholder figure.

5 Intra-Case State

Placeholder.

5.1 Feature Extraction

Placeholder.

5.2 Dimensionality Reduction

Placeholder.

5.3 State Clustering

Placeholder.

5.4 Drift Signals

Placeholder.

6 Resource State

Placeholder.

6.1 Feature Extraction

Placeholder.

6.2 Dimensionality Reduction

Placeholder.

6.3 State Clustering

Placeholder.

6.4 Drift Signals

Placeholder.

7 Inter-Case State

Placeholder.

7.1 Feature Extraction

Placeholder.

7.2 Dimensionality Reduction

Placeholder.

7.3 State Clustering

Placeholder.

7.4 Drift Signals

Placeholder.

8 Implementation

Placeholder.

8.1 Kairo

Placeholder.

8.2 Dashboard

Placeholder.

9 Evaluation

In this chapter, you present your evaluation. You should have extensively discussed with your supervisor what you are evaluating and why ...

Some typical sections that you need here:

9.1 Experimental Setup

Describe the setup of your experiments. If you apply a pipeline of different techniques, show the pipeline. Present the parameters of your experiments in a table, and, describe them. Typical Elements:

- Input Data Set(s) Used
- Algorithm(s) Used
- Parameter(s) Used
- ...

9.2 Results

Show the results. Try to present your results as structured as you can. A good result (sub)section, follows the following structure:

- Describe what results you are going to Show
- Present an initial hypothesis about the results, e.g., We expect the quality metric M to behave like this, conditional to this parameter P.
- Show the Results
- Confirm the hypothesis.
- If there are results that are not according to the hypothesis, you have to be able to explain why this is the case!!!

9.3 Threats to Validity

Here you discuss any element of your experiments (usually depending on the setup), e.g., data sets used, parameters assessed, that might have implications for the validity of your results. For example, if you have not considered a specific range of parameter values, it is unclear how the algorithm will behave for these settings. If you have excluded certain data, this might have its implications for the generalizability of your results. In a way, this is the discussion section of your thesis.

10 Discussion

In this section, you discuss your general approach. What design decisions did you make, and, what are the implications of this? Maybe your approach computes an exact intermediary result, which is costly in terms of time. Does a suitable approximation approach exist? What are the implications for that approach. You can also dive deeper in very related work and explain why it does not work in certain cases.

11 Conclusion

Your conclusion actually follows the same structure as your abstract. However, you present what the reader has seen, again, using the same concepts as you have already used in your abstract.

Future Work. Furthermore, you present interesting directions for future work!

A Some additional stuff

Try to avoid appendices. If you can't, this is a place to show all kinds of results that are supportive, yet, not critical for your thesis.

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